

CADT

Course: Database Administration

Lecturer: Mr. Thear Sophal

Week 3 – Practical Session

User Grant and Access Part I

✓ TODOs

Part 1: Database Design and Data Insertion

1. Database Creation

- o Design a relational database named **banking_system**.
- o The schema should include the following tables:
 - Customers
 - Accounts
 - Transactions
 - Loans

2. Database Creation Method

- o Use a Generative AI tool (e.g., ChatGPT) to generate SQL DDL statements, or
- o Create the tables manually using SQL.

3. Insert Sample Data

- o Populate each table with a minimum of **20 meaningful records**.
- o You may generate data using AI tools or insert it manually.

Part 2: User Creation and Privilege Management

Scenario: You are the Database Administration Manager for the **banking_system** database. You are responsible for creating user accounts and assigning appropriate permissions to ensure secure and efficient database usage.

A. Database Administration Team – you have 3 subordinates

1. Michael - Deputy Manager

- o Username : michael
- o Password : michael123\$
- o Privileges : ALL PRIVILEGES on all tables + GRANT OPTION

2. Jessica - Database Administrator

- o Username : jessica
- o Password : jessica2023\$
- o Initial Privileges: ALTER, CREATE VIEW, DELETE, SELECT, INSERT, UPDATE on all tables
 - Follow-up: Revoke the above privileges + Grant ALL PRIVILEGES on all tables

3. James - Database Administrator

- o Username : james
- o Password : james2024\$
- o Privileges : ALTER, CREATE VIEW, DELETE, SELECT, INSERT, UPDATE on all tables

B. W

eb Development Team – 5 members

1. David - Web Development Manager

- o Username : david
- o Password : david123456\$
- o Privileges : ALTER, CREATE VIEW, INSERT, SELECT, UPDATE on all tables

2. Matthew - Senior Developer

- o Username : matthew
- o Password : matthew6789\$
- o Privileges : CREATE VIEW, SELECT, INSERT, UPDATE on all tables

3. Emily - Junior Developer

- o Username : emily
- o Password : emily1234\$
- o Privileges : SELECT, INSERT, UPDATE on the following tables: Customers, Accounts, Transactions

4. John - Junior Developer

- o Username : john
- o Password : john1234\$
- o Privileges : SELECT, INSERT, UPDATE on the following tables: Customers, Accounts Transactions

5. Mario - Intern

- o Username : mario
- o Password : mario1234\$
- o Privileges : SELECT, INSERT, UPDATE on the following tables: Customers, Accounts Transactions for 100 days

How to Submit?

Deliverables to submit in UMS

- **SQL script for:**
 - o Database and table creation
 - o Inserting sample data
 - o Creating user accounts
 - o Granting and revoking privileges
- **Documentation or screenshots that demonstrate:**
 - o Privileges assigned to each user
 - o Verification of operations allowed or denied based on user roles